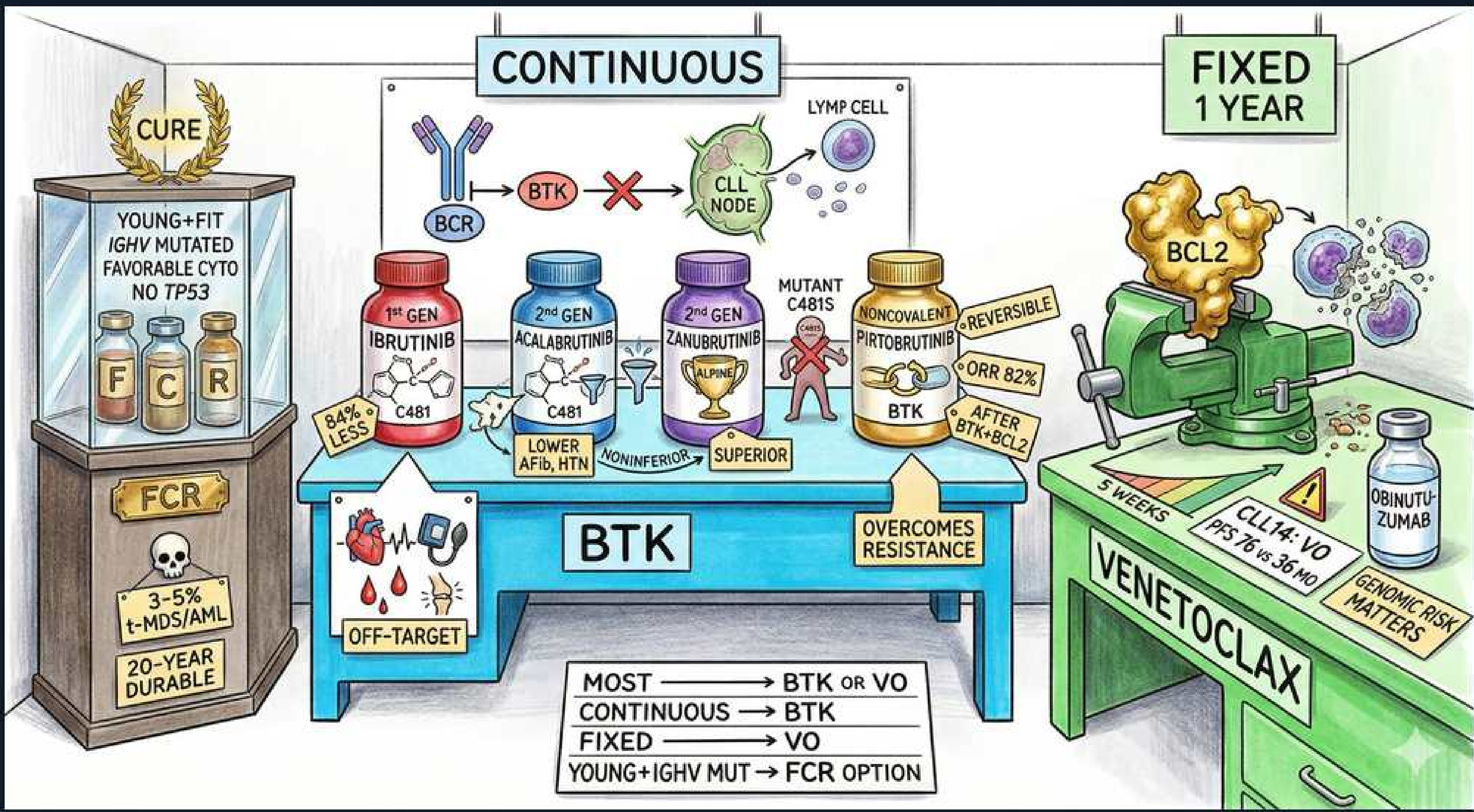
CLL — Treatment: BTK Inhibitors, Venetoclax & FCR



1. FCR cure niche

WHAT YOU SEE

Trophy case CURE, young fit IGHV mutated no TP53

WHAT IT ENCODES

FCR (fludarabine + cyclophosphamide + rituximab) is the only CLL regimen with a survival-curve plateau suggesting functional cure, but ONLY in young (<65), fit, IGHV-MUTATED patients WITHOUT del(17p)/TP53 mutation. In this narrow niche, mutated patients can achieve very long, treatment-free remissions.

BOARD TRAP

Never give FCR if del(17p)/TP53-mutated (chemoresistant) OR IGHV-unmutated (no plateau, short remission). In the targeted-therapy era many centers have abandoned FCR even for the eligible niche, given t-MDS/AML risk.

---

2. FCR risk: t-MDS/AML

WHAT YOU SEE

Skull box FCR, 3-5% t-MDS/AML

WHAT IT ENCODES

Chemoimmunotherapy (especially fludarabine-based FCR) carries a 3-5% risk of therapy-related MDS/AML and prolonged cytopenias/immunosuppression. Fludarabine is also a strong T-cell suppressant raising opportunistic infection risk.

BOARD TRAP

This long-term secondary-leukemia risk, plus equal/superior efficacy of targeted agents, is the main reason BTKi and venetoclax have displaced chemo as frontline for most patients.

---

3. Continuous BTK

WHAT YOU SEE

CONTINUOUS banner over BTK table

WHAT IT ENCODES

BTK inhibitors are given CONTINUOUSLY — daily until disease progression or unacceptable toxicity. They control disease without typically achieving deep MRD-negative remissions when used alone, hence indefinite dosing.

BOARD TRAP

BTKi = continuous/indefinite. Venetoclax-based regimens are FIXED duration (1 year with obinutuzumab). Mixing these up is a classic exam error — match the agent to its dosing schedule.

---

4. BCR→BTK→CLL node

WHAT YOU SEE

BCR receptor, BTK, X blocking CLL node

WHAT IT ENCODES

Bruton tyrosine kinase (BTK) transmits B-cell receptor (BCR) survival signals. BTK inhibitors block this signaling, evicting CLL cells from protective lymph-node niches into the blood — causing a characteristic transient treatment-related lymphocytosis that is NOT progression.

BOARD TRAP

Early redistribution lymphocytosis on a BTKi is expected and benign — do NOT mistake the rising lymphocyte count in the first weeks/months for treatment failure.

---

5. Ibrutinib 1st gen

WHAT YOU SEE

Red bottle 1st GEN IBRUTINIB, C481

WHAT IT ENCODES

Ibrutinib is the first-generation covalent (irreversible) BTK inhibitor, binding cysteine-481 (C481) in the ATP-binding pocket. It was the first to revolutionize CLL therapy and works across high-risk groups including del(17p).

BOARD TRAP

Ibrutinib has more OFF-TARGET inhibition (EGFR, ITK, TEC, etc.) than 2nd-gen agents, driving its higher rate of cardiac and bleeding toxicities.

---

6. Ibrutinib off-target

WHAT YOU SEE

Heart/blood OFF-TARGET icons

WHAT IT ENCODES

Ibrutinib toxicities (largely off-target): atrial fibrillation/flutter, hypertension, bleeding (it impairs platelet function — hold around surgery), and arthralgias/diarrhea. AFib management is complicated because anticoagulation adds to the bleeding risk.

BOARD TRAP

AFib + bleeding/HTN are the classic ibrutinib board adverse effects. Avoid combining with warfarin; be cautious with anticoagulation generally, and hold periprocedurally for bleeding risk.

---

7. Acalabrutinib 2nd gen

WHAT YOU SEE

Blue bottle 2nd GEN ACALABRUTINIB, lower AFib/HTN

WHAT IT ENCODES

Acalabrutinib is a more selective 2nd-generation covalent BTKi (also binds C481) with less off-target activity → significantly lower rates of AFib, hypertension, and bleeding than ibrutinib.

BOARD TRAP

Headache is the characteristic acalabrutinib side effect (often transient, caffeine-responsive). Note acalabrutinib absorption is pH-dependent — avoid PPIs (separate from antacids/H2 blockers).

---

8. Zanubrutinib superior

WHAT YOU SEE

Green bottle ZANUBRUTINIB, trophy SUPERIOR

WHAT IT ENCODES

Zanubrutinib is a 2nd-generation covalent BTKi that demonstrated SUPERIOR progression-free survival and a better cardiac-safety profile (less AFib) than ibrutinib in head-to-head trials (ALPINE), making it a preferred BTKi.

BOARD TRAP

'Superior' here is head-to-head vs ibrutinib — both 2nd-gen agents (acala, zanu) beat ibrutinib on tolerability; zanubrutinib additionally beat it on efficacy.

9. Pirtobrutinib C481S

WHAT YOU SEE

Pirtobrutinib, MUTANT C481S, NONCOVALENT, reversible

WHAT IT ENCODES

Pirtobrutinib is a NONCOVALENT, REVERSIBLE BTK inhibitor that does not depend on the C481 cysteine — so it retains activity against the C481S resistance mutation that arises after covalent BTKi exposure. Used after prior covalent BTKi (and often BCL2) failure.

BOARD TRAP

C481S mutation = acquired resistance to COVALENT BTKi (ibrutinib/acala/zanu); pirtobrutinib overcomes it precisely because it binds noncovalently and not at C481.

10. Venetoclax BCL2

WHAT YOU SEE

BCL2 clamp on vise, VENETOCLAX bench

WHAT IT ENCODES

Venetoclax is a BCL2 inhibitor (BH3-mimetic) that restores apoptosis in CLL cells, which depend on anti-apoptotic BCL2. Given as FIXED-duration therapy — classically 12 months combined with obinutuzumab (CLL14 regimen, 'VO') — it achieves deep, often MRD-undetectable remissions.

BOARD TRAP

Venetoclax is fixed-duration (contrast with continuous BTKi). It produces deeper MRD-negative responses than BTKi monotherapy, enabling time-limited treatment.

11. TLS risk ramp-up

WHAT YOU SEE

WEEKS dose escalation, CLL14 VO obinutuzumab

WHAT IT ENCODES

Venetoclax causes TUMOR LYSIS SYNDROME (hyperkalemia, hyperphosphatemia, hyperuricemia, hypocalcemia → AKI/arrhythmia). It is started at a low dose with a mandatory weekly 5-step ramp-up (20→50→100→200→400 mg) plus hydration, allopurinol/rasburicase, and lab monitoring.

BOARD TRAP

TLS is THE venetoclax board hazard. High tumor burden (bulky nodes ≥5 cm or high ALC) and/or reduced renal function = high TLS risk → hospitalize for the first doses. Never start full-dose venetoclax.

12. Obinutuzumab CD20

WHAT YOU SEE

Small OBINUTUZUMAB vial on the green Venetoclax bench beside a GENOMIC RISK MATTERS tag — the anti-CD20 partner that debulks tumor before the venetoclax ramp

WHAT IT ENCODES

Obinutuzumab is a type-II glycoengineered anti-CD20 monoclonal antibody partnered with venetoclax in fixed-duration frontline therapy (CLL14). It adds depth of response; obinutuzumab debulks tumor before venetoclax escalation, helping mitigate TLS.

BOARD TRAP

Anti-CD20 antibodies (obinutuzumab, rituximab) carry infusion reactions and risk of hepatitis B reactivation — screen HBV before starting, and watch for first-dose cytokine-release/infusion reactions (worse with obinutuzumab).

13. Richter transformation

WHAT YOU SEE

Cluster of large angry transformed lymphocytes lurking behind the BTK table — CLL morphing into an aggressive DLBCL node

WHAT IT ENCODES

Richter transformation is the change of CLL into an aggressive lymphoma — most often diffuse large B-cell lymphoma (DLBCL). Suspect it with rapidly enlarging asymmetric nodes, B symptoms, sharply rising LDH, and PET hot-spots (high SUV). Diagnosis requires biopsy; prognosis is poor.

BOARD TRAP

Richter → DLBCL (occasionally Hodgkin lymphoma). A clonally-related DLBCL arising from the CLL has a far worse prognosis than de novo DLBCL. A single intensely PET-avid node in a CLL patient should be biopsied, not assumed to be progressive CLL.

14. Algorithm box

WHAT YOU SEE

Bottom rules: MOST→BTK or VO, FIXED→VO, young IGHV mut→FCR

WHAT IT ENCODES

Frontline algorithm: MOST patients → a 2nd-gen BTKi (acala/zanu, continuous) OR venetoclax+obinutuzumab (VO, fixed 1 year). Choose VO when fixed-duration is desired; choose BTKi for continuous oral therapy avoiding TLS ramp-up. FCR only for young, fit, IGHV-mutated, no-TP53 patients. del(17p)/TP53 → BTKi or venetoclax, never chemo.

BOARD TRAP

Patient/comorbidity factors steer the choice: cardiac disease/AFib or anticoagulation favors VO over BTKi; poor renal function or very high tumor burden (TLS risk) and a preference to avoid daily lifelong pills favors BTKi over venetoclax.